

Assignment 3: TCP Chat Server and Client

1. General Description

In this assignment, we implemented a multi-user chat application using the TCP protocol in C. The solution consists of a server that manages connections and distributes messages, and a client that allows users to send and receive messages simultaneously. We used the pthread library to handle multiple clients and tasks at the same time.

2. Server Implementation (hw3server.c)

The server uses a **"Thread-per-Client"** model.

- **Connection Handling:** The server listens on a port specified in the command line. When a new client connects, the server accepts the connection and creates a new thread (handle_client) dedicated to that client.
- **Client Management:** We use a global array to store all connected clients. To prevent data conflicts (race conditions) when multiple threads access this list, we use a Mutex (clients_mutex).
- **Message Processing:**
 - **Broadcast:** Normal messages are sent to all connected clients.
 - **Whisper (Private):** If a message starts with @, the server parses the target name. It searches for the client with that name in the list and sends the message only to them.
- **Disconnection:** When a client disconnects, the server cleans up the resources and removes the client from the global list.

3. Client Implementation (hw3client.c)

The client is designed to handle sending and receiving messages at the same time without blocking. It uses two main threads:

- **Main Thread (Sender):** This thread reads input from the user's keyboard. It checks for the !exit command to quit safely or sends the text to the server.
- **Receiver Thread:** This thread runs in the background. It constantly listens for incoming messages from the server and prints them to the screen immediately. This ensures the user can see messages from others even while typing.

4. Submitted Files

- hw3server.c: The server source code.
- hw3client.c: The client source code.
- hw3.h: Header file containing common constants (like buffer size) and libraries.
- Makefile: A script to compile the project using gcc with the -pthread flag.